

Notes: Liang, Lu, Mu & Okumura (JPE, 2026)
Algorithm Design: A Fairness-Accuracy Frontier

Contents

1	Big picture	3
2	Data and measurement: model primitives, empirical datasets, and constructed objects	3
2.1	Primitive objects: covariates, types, groups, and algorithms	3
2.2	Feasible set, Pareto frontier, and fairness-accuracy frontier	3
2.3	Fairness-accuracy preferences	4
2.4	Group optimal points, fairness optimal point, and group skew	4
2.5	Empirical health-care datasets	4
2.6	Empirical estimation objects	5
3	Models and theoretical specifications	5
3.1	Group errors and feasible set	5
3.2	FA dominance	5
3.3	Theorem 1: geometric characterization of the frontier	5
3.4	Algorithm sequence implementing the frontier	5
3.5	Group-specific algorithms	6
3.6	Input design and garbling	6
3.7	When input design is without loss	6
3.8	Banning group identity and other covariates	6
4	Identification strategy and empirical implementation	6
4.1	Conceptual identification	6
4.2	Testing group skew and group balance	7
4.3	Estimating feasible sets	7
4.4	Limits	7
5	Tables and figures	7
5.1	Figure 1: The fairness-accuracy frontier	7
5.2	Figure 2: Example feasible set and fairness-accuracy frontier for group-balanced and r-skewed inputs	8
5.3	Figure 3: Algorithms that implement the FA frontier	8
5.4	Figure 4: Group-specific algorithms and Figure 5: Special cases	8
5.5	Figure 6: Input-design fairness-accuracy frontier and Figure 7: Excluding a covariate	8
5.6	Figure 8: Group identity and the value of additional covariates	8
5.7	Figure 9: Cross-validated group-optimal points and Table 1: Test outcomes	9
5.8	Figure 10: Estimated feasible set for the class of linear algorithms	9

5.9	Appendix Figure 11: The boundary K_X and Figure 12: Classification and surrogate loss . . .	9
6	Interpretive synthesis	11
7	Short summary	11
8	Conclusion	11
8.1	Core conclusion	11
8.2	Economic intuition	11
8.3	Policy takeaway	11
8.4	Final takeaway	12

1 Big picture

- **Question.** How does the information available to an algorithm determine the trade-off between predictive accuracy and group fairness?
- **Object.** The paper defines a **fairness-accuracy frontier**: the set of feasible group-error pairs that cannot be improved in both accuracy and fairness.
- **Core condition.** The shape of the frontier is governed by whether inputs are **group balanced** or **group skewed**. Group skew means one group remains advantaged even when the algorithm is optimized for the other group.
- **Main theorem.** If inputs are group balanced, the fairness-accuracy frontier is exactly the standard Pareto frontier. If inputs are group skewed, the frontier includes an additional positively sloped segment where fairness improves only because both groups' errors rise.
- **Algorithmic characterization.** Moving along the frontier corresponds to switching decisions at covariate realizations ordered by their relative marginal impact on the two groups' errors.
- **Information design.** A policymaker may regulate inputs while an accuracy-maximizing agent chooses the algorithm. Under endpoint conditions, garbling the inputs can implement the same outcome as direct algorithm control.
- **Policy message.** Banning group identity or other informative variables is not generically pro-fairness. When group identity is available, banning any decision-relevant covariate is strictly suboptimal for all fairness-accuracy designers.
- **Empirical illustration.** In two health-care datasets, one frontier is group skewed and displays strong fairness-accuracy conflict, while the other is approximately group balanced and displays almost no fairness-accuracy conflict.

2 Data and measurement: model primitives, empirical datasets, and constructed objects

2.1 Primitive objects: covariates, types, groups, and algorithms

Individuals have covariates X , outcomes or types Y , and group identity $G \in \{r, b\}$. An algorithm maps observed covariates into a decision d . A loss function $\ell(d, Y)$ measures decision error or harm. The expected loss for group g is the group error

$$e_g(a) = \mathbb{E}[\ell(a(X), Y) \mid G = g].$$

Thus every algorithm is summarized by a two-dimensional error vector $e(a) = (e_r(a), e_b(a))$.

Interpretation: The paper intentionally studies group-level error pairs rather than a single aggregate loss because fairness concerns are about how algorithmic errors are distributed across groups.

2.2 Feasible set, Pareto frontier, and fairness-accuracy frontier

The feasible set given covariates X is

$$E_X = \{e(a) : a \in \Delta(A)\},$$

where randomization over algorithms makes the feasible set convex. The Pareto frontier is

$$P_X = \{e \in E_X : \text{no } e' \in E_X \text{ satisfies } e' \succ_{PD} e\},$$

and the fairness-accuracy frontier is

$$F_X = \{e \in E_X : \text{no } e' \in E_X \text{ satisfies } e' \succ_{FA} e\}.$$

Interpretation: The FA frontier removes error pairs that are dominated both in accuracy and disparity. It is the set of potentially defensible outcomes for some fairness-aware designer.

2.3 Fairness-accuracy preferences

The paper considers a broad class of preferences. FA dominance requires lower error for both groups and a smaller difference $|e_r - e_b|$. This class includes:

- **Utilitarian** designers who minimize weighted aggregate error.
- **Rawlsian** designers who minimize the worse group error $\max\{e_r, e_b\}$.
- **Egalitarian** designers who minimize $|e_r - e_b|$.
- **Constrained optimization** designers who impose $|e_r - e_b| \leq c$ before minimizing aggregate error.

Interpretation: This is why the results are robust: the frontier is not tied to a single fairness objective.

2.4 Group optimal points, fairness optimal point, and group skew

The group-optimal points are

$$r_X = \arg \min_{e \in E_X} e_r, \quad b_X = \arg \min_{e \in E_X} e_b.$$

The fairness-optimal point is

$$f_X = \arg \min_{e \in E_X} |e_r - e_b|.$$

The covariate vector X is r -skewed if group r has lower error than group b at both r_X and b_X ; it is b -skewed symmetrically; otherwise it is group balanced.

Interpretation: Group skew is a geometric property of the feasible set. It captures whether the information environment inherently favors one group.

2.5 Empirical health-care datasets

The empirical section uses two datasets.

- **Obermeyer et al. (2019).** 48,784 patient observations. Groups are Black and White patients. The outcome is whether true health needs exceed the 97th percentile. Covariates include demographics, chronic illness indicators, health-care cost variables, biomarkers, and medication variables.
- **Strack et al. (2014).** 101,766 diabetes-care observations. Groups are women and men. The outcome is hospital readmission after release. Covariates include age, diabetic medications, inpatient days, and related clinical variables.
- In both datasets the loss is misclassification loss, $\ell(d, y) = 1\{d \neq y\}$.

Interpretation: These data are used to show that the theory’s objects—group balance, group skew, feasible sets, and FA frontiers—can be estimated in real algorithmic fairness applications.

2.6 Empirical estimation objects

Group-optimal points are estimated by fivefold cross-validation. For unconstrained algorithms the paper uses random forests; for linear algorithms it uses logistic regression for group-optimal classifiers and a hinge-loss linear-programming relaxation to trace the feasible frontier.

Interpretation: The empirical analysis operationalizes the theory: estimate group-optimal points, classify the input environment as group balanced or skewed, and then trace the feasible frontier.

3 Models and theoretical specifications

3.1 Group errors and feasible set

$$e_g(a) = \mathbb{E}[\ell(a(X), Y) \mid G = g], \quad e(a) = (e_r(a), e_b(a)).$$

$$E_X = \{e(a) : a \in \Delta(A)\}.$$

Interpretation: The feasible set is the algorithmic opportunity set in error space.

3.2 FA dominance

An error pair e' FA-dominates e if it has weakly lower errors for both groups and weakly smaller error disparity, with at least one strict improvement:

$$e' \succ_{FA} e \quad \Rightarrow \quad e'_r \leq e_r, \quad e'_b \leq e_b, \quad |e'_r - e'_b| \leq |e_r - e_b|.$$

Interpretation: FA dominance is stricter than Pareto dominance because it adds the fairness criterion.

3.3 Theorem 1: geometric characterization of the frontier

- If X is group balanced, then $F_X = P_X$.
- If X is g -skewed, then F_X is the boundary of E_X between g_X and f_X that contains P_X .

Interpretation: This theorem is the core of the paper. A positive fairness-accuracy conflict, where fairness improves only by increasing both groups' errors, arises exactly when the inputs are group skewed.

3.4 Algorithm sequence implementing the frontier

For each covariate realization x , define

$$\Delta_x^g = P(X = x \mid G = g) \mathbb{E}[\ell(1, Y) - \ell(0, Y) \mid X = x, G = g],$$

and

$$h(x) = \begin{cases} \Delta_x^b / \Delta_x^r, & \Delta_x^r \neq 0, \\ \infty, & \Delta_x^r = 0. \end{cases}$$

Ordering covariates by $h(x)$, the frontier is implemented by starting with the r -optimal algorithm and switching decisions at successive covariate values.

Interpretation: This shows that the frontier has a concrete algorithmic implementation: moving along it is not an abstract optimization artifact but a sequence of decision-rule changes.

3.5 Group-specific algorithms

If algorithms can condition on group identity, the feasible set is a rectangle and the FA frontier is the line from $r_X = b_X$ to f_X .

Interpretation: Group-specific algorithms can hold the worse-off group at its minimal error and adjust the advantaged group's error to improve fairness.

3.6 Input design and garbling

In the information-design part, the designer chooses a garbling of the primitive inputs, and an agent then chooses an accuracy-maximizing algorithm. Let H denote the error pairs that the agent weakly prefers to the no-information benchmark.

Interpretation: This models legal or technological input regulation: the policymaker controls what information the algorithm can use, not necessarily the algorithm itself.

3.7 When input design is without loss

For group-balanced inputs, it is enough to check whether r_X and b_X lie in H . For group-skewed inputs, it is enough to check the advantaged group-optimal point and the fairness-optimal point f_X .

Interpretation: If the relevant endpoints are acceptable to the agent, a designer can use garbling to induce the agent to select the designer's preferred outcome.

3.8 Banning group identity and other covariates

If X is strictly group balanced, adding group identity strictly improves the input-design frontier for every fairness-accuracy designer. If group identity is available, any decision-relevant additional covariate strictly improves the frontier.

Interpretation: The result is a warning against simplistic bans. In some environments, group identity and other informative variables help designers implement fairer outcomes.

4 Identification strategy and empirical implementation

4.1 Conceptual identification

The frontier is identified from the joint distribution of (X, Y, G) , the loss function, and the permitted algorithm class A . Once these primitives are fixed, E_X , P_X , F_X , r_X , b_X , and f_X are well-defined. Group balance or skew is determined by the locations of r_X and b_X relative to the 45-degree line.

Interpretation: This is not a causal-treatment design; it is a structural/geometric characterization of what algorithms can achieve given an information environment.

4.2 Testing group skew and group balance

For the Obermeyer data, the paper tests whether the covariates are White-skewed. For the Strack diabetes data, it tests approximate group balance using a tolerance $\delta = 0.01$. The procedure uses five random train-test splits, bootstrap p-values, and the median-p-value rule of Auerbach et al. (2024).

Interpretation: The tests are designed directly around the theory’s geometry: do the group-optimal points lie on the same side of the 45-degree line or nearly on it?

4.3 Estimating feasible sets

For linear classifiers $a(x) = 1\{x^\top \beta \geq 0\}$, the paper approximates extreme points by solving weighted error-minimization problems for many (α_r, α_b) . A mixed-integer linear program is relaxed to a hinge-loss linear program, evaluated by fivefold cross-validation, and then convexified.

Interpretation: The empirical output is an estimated frontier, not a single coefficient.

4.4 Limits

The estimates depend on the algorithm class, loss function, group definition, and sample. Linear algorithms may understate the full nonlinear feasible set, and symmetric misclassification loss may not be the correct welfare loss in every application.

Interpretation: The empirical section is an illustration of the framework, not a universal statement about all medical algorithms.

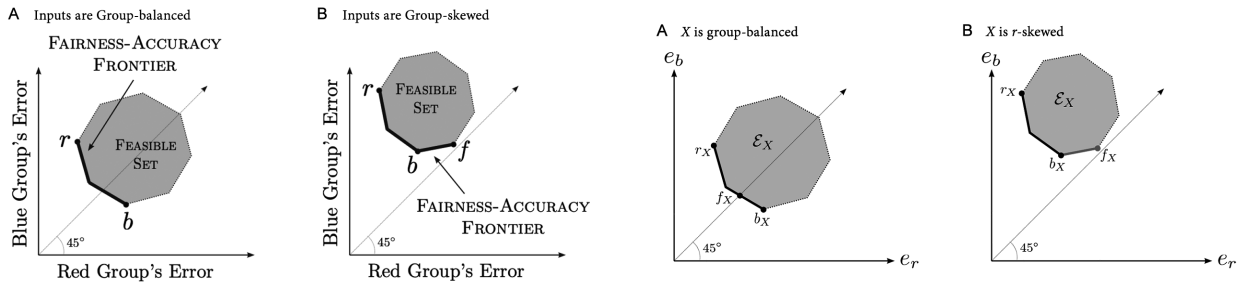
5 Tables and figures

5.1 Figure 1: The fairness-accuracy frontier

Read: Panel A shows group-balanced inputs, where the FA frontier is the Pareto frontier. Panel B shows group-skewed inputs, where the frontier includes a positively sloped segment from b to f .

Interpretation: The figure displays the central distinction: under group skew, fairness can require raising both groups’ errors.

Economic message: The figure gives the paper’s main economic message in one picture.



5.2 Figure 2: Example feasible set and fairness-accuracy frontier for group-balanced and r-skewed inputs

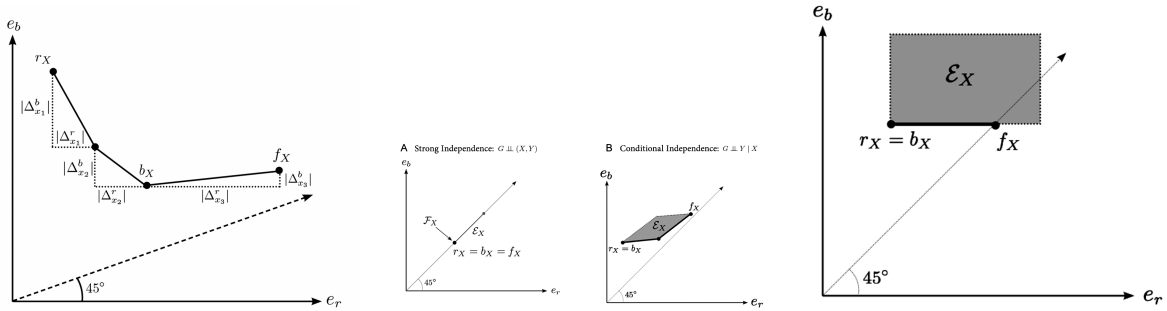
Read: Panel A has group-optimal points on opposite sides of the 45-degree line. Panel B has both group-optimal points on the same side.

Interpretation: Figure 2 formalizes group balance and group skew as properties of the feasible set.

5.3 Figure 3: Algorithms that implement the FA frontier

Read: The figure traces algorithms a_0, a_1, a_2, a_3 generated by sequentially switching decisions at covariate realizations.

Interpretation: Moving along the frontier corresponds to concrete changes in the algorithm's decision rule.



5.4 Figure 4: Group-specific algorithms and Figure 5: Special cases

Read: Figure 4 shows that group-specific algorithms make the feasible set rectangular. Figure 5 shows strong independence and conditional independence special cases.

Interpretation: These figures clarify when group-specific algorithms or independence assumptions simplify the frontier.

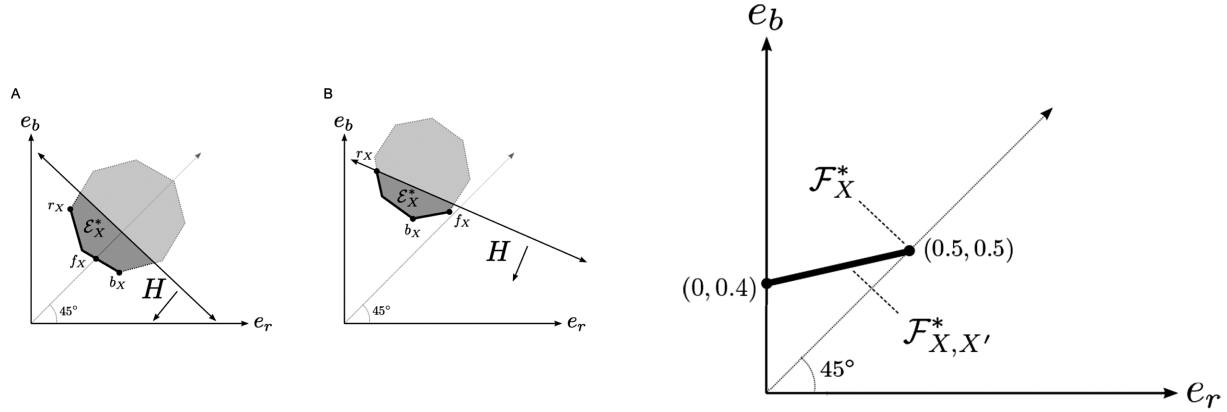
5.5 Figure 6: Input-design fairness-accuracy frontier and Figure 7: Excluding a covariate

Read: Figure 6 shows when input design can or cannot implement the unconstrained frontier. Figure 7 gives an example where excluding a covariate is strictly optimal for the designer.

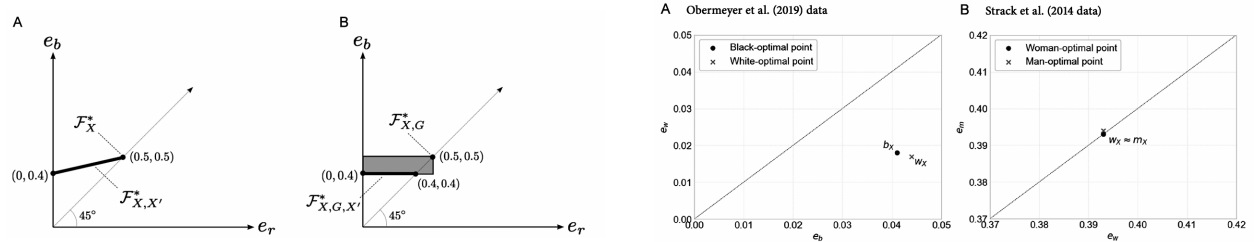
Interpretation: Input design can be powerful, but only when the agent's accuracy objective is sufficiently aligned with the designer's frontier endpoint.

5.6 Figure 8: Group identity and the value of additional covariates

Read: Without group identity, adding X' can be harmful for a fairness-motivated designer. With group identity, the designer can garble X' differently across groups and achieve a superior point.



Interpretation: Group identity can increase, rather than reduce, the value of other decision-relevant information.



5.7 Figure 9: Cross-validated group-optimal points and Table 1: Test outcomes

Read: Figure 9 estimates group-optimal points. Table 1 classifies Dataset 1 as White-skewed and Dataset 2 as group balanced for both algorithm specifications.

Interpretation: The empirical classification matters because Theorem 1 says the fairness interpretation changes sharply between group skew and group balance.

5.8 Figure 10: Estimated feasible set for the class of linear algorithms

Read: The figure plots estimated feasible sets for the two datasets and shows how the frontiers change when group identity is available.

Interpretation: For Obermeyer, some fairness-motivated designers cannot induce their preferred point through input design. For Strack, input design is effectively without loss.

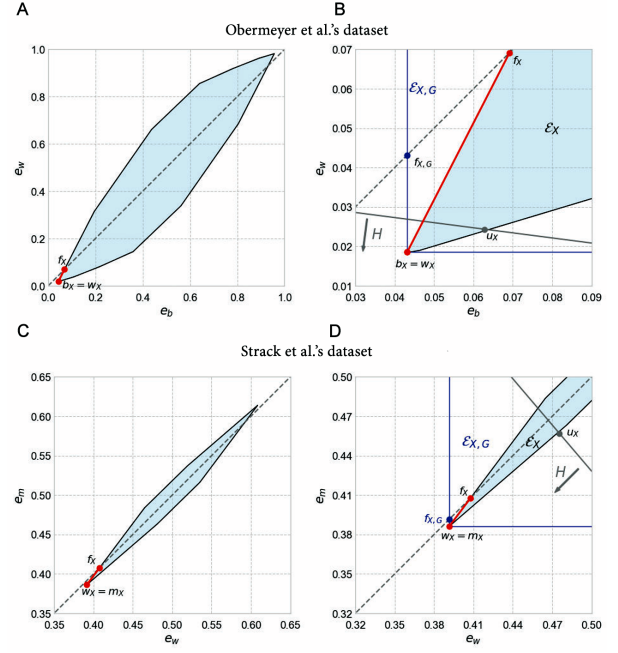
5.9 Appendix Figure 11: The boundary K_X and Figure 12: Classification and surrogate loss

Read: Figure 11 illustrates a proof object for the boundary of the feasible set. Figure 12 illustrates the hinge surrogate used in the LP relaxation.

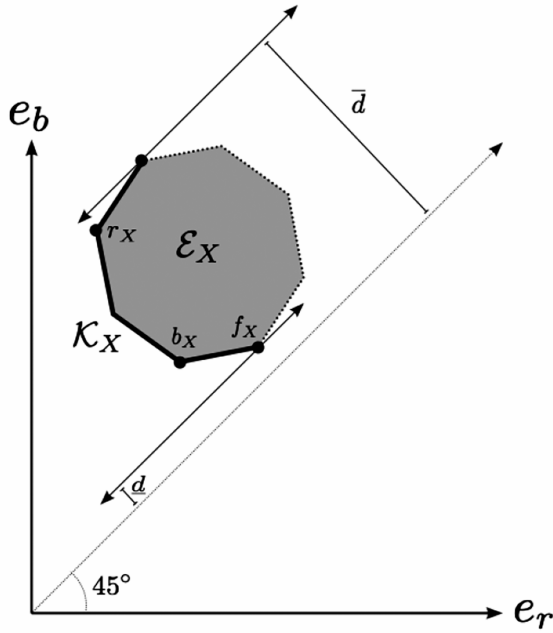
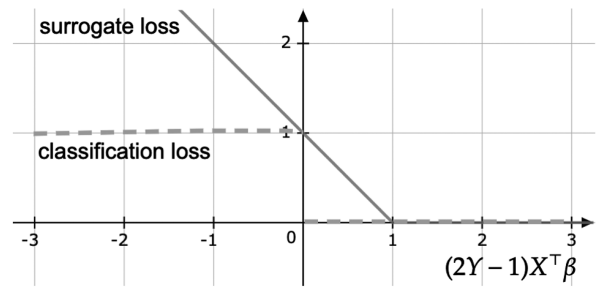
TABLE 1
TEST OUTCOMES

	Dataset 1		Dataset 2	
	Group Balance/ Skew	Median p -Value	Group Balance/ Skew	Median p -Value
Unconstrained	w skewed	<.0001	Group balanced	.0437
Linear	w skewed	<.0001	Group balanced	<.0001

NOTE.—We report whether each dataset is group balanced or group skewed for the two specifications of A . Reported p -values are computed via bootstrap with size 10,000.



Interpretation: These figures connect the geometric proof strategy to the computational implementation of the empirical frontier.

(a) Boundary K_X 

(b) Classification and surrogate loss

Appendix Figures 11 and 12

6 Interpretive synthesis

- **The frontier is a diagnostic tool.** It identifies which fairness interventions are potentially defensible and which are dominated.
- **Group skew is the key primitive.** Fairness can require sacrificing both groups' accuracy only when inputs inherently advantage one group.
- **Input bans are context dependent.** Bans can be useful when an agent's incentives are misaligned and group identity is not available, but can be strictly harmful when group identity enables group-dependent garbling.
- **Empirical estimation matters.** The two health-care datasets produce different frontier shapes, so one-size-fits-all algorithmic fairness rules are not justified.

7 Short summary

- The paper defines the fairness-accuracy frontier for group errors.
- The frontier's shape depends on whether inputs are group balanced or group skewed.
- Group balance implies the FA frontier equals the Pareto frontier.
- Group skew adds a segment where fairness improves only by increasing both groups' errors.
- The frontier can be implemented by ordered changes in algorithmic decisions.
- Input design can implement the designer's preferred outcome when the relevant endpoints are acceptable to the agent.
- Banning group identity is strictly suboptimal under strict group balance.
- Once group identity is available, banning a decision-relevant covariate is strictly suboptimal.
- Empirically, the Obermeyer data are White-skewed and the Strack diabetes data are approximately group balanced.

8 Conclusion

8.1 Core conclusion

The paper shows that the fairness-accuracy trade-off is governed by group skew. If inputs are group balanced, fairness does not justify Pareto-dominated error pairs. If inputs are group skewed, reducing disparity can require raising both groups' errors.

8.2 Economic intuition

An algorithm's fairness problem is a geometry problem. The feasible set tells which combinations of group errors can be achieved. If one group is inherently easier to predict, then equalizing errors can require degrading predictions for the advantaged group and sometimes for both groups.

8.3 Policy takeaway

The right information policy depends on the estimated frontier. Banning sensitive variables or test scores is not always the fairness-enhancing move; in some cases, group-dependent garbling using group identity is the more powerful fairness tool.

8.4 Final takeaway

The paper reframes algorithmic fairness from a debate over particular variables into a frontier problem: estimate the feasible fairness-accuracy frontier, classify the input environment, and then choose the point or garbling that best matches the designer's preference.